TEXORA · LECTURE SHEET

Garments Inspection and Quality Control

The layered inspection system of a garment factory — in-line, end-of-line, measurement audit, final AQL audit, needle detection and carton audit — with the measurement-check calculation worked in full.

AM-4101 · ADVANCED APPAREL MANUFACTURING-I · TOPIC 4 · SPECIALIZATION — APPAREL MANUFACTURING

1. Inspection is a system of layers

No single inspection can carry a garment's quality. Factories layer checks so that each defect is caught at the cheapest point where it can still be corrected: **in-line inspection** patrols operations while sewing; **end-of-line inspection** checks every finished garment; **measurement audit** verifies dimensions on a sample; the **final audit** samples packed goods statistically and decides shipment (AM-3207 Topic 9); and safety gates — **needle/metal detection** and **carton audit** — protect the customer and the count. This topic organises the layers; the statistical engine inside the final audit is AM-3207 Topic 9, built on TE-3107.



The inspection chain. A failure at any gate returns the garment (or the lot) for correction before it moves right.

Figure 1 — Inspection layers in production order. Metal detection comes after all sewing, trimming and pressing, so nothing metallic can enter after the gate.

2. In-line control and the traffic-light idea

In-line inspectors check work at the operation, where a fault costs seconds to fix instead of a rejected garment. Lines compute DHU at the in-line points (AM-3207 Topic 9): a roving check of 250 garments finding 15 defects gives DHU = 15 ÷ 250 × 100 = **6.0**. Many factories colour-code stations by recent DHU — green running clean, amber watched, red stopped and diagnosed — so the calculation becomes a floor signal any supervisor can read at a glance. End-of-line then checks 100% of garments visually

and by critical measurements; its repairs loop back to the responsible operation with the defect identified, which is how inspection data becomes process improvement rather than merely sorting.

3. Measurement audit — worked example

Appearance can pass while dimensions fail, so auditors measure sampled garments at the specification's points of measurement. For the course shirt the chest specification is **112 cm ± 1.5 cm** (acceptable range 110.5–113.5 cm) for size M.

> **WORKED EXAMPLE — TEN MEASURED CHESTS**
>
> Measured (cm): 112.0 · 111.5 · 112.4 · **110.2** · 112.8 · 111.9 · 112.1 · **113.8** · 112.3 · 111.6.
>
> 110.2 is below 110.5 — out of tolerance. 113.8 is above 113.5 — out of tolerance. **2 of 10 garments (20%) fail measurement.**
>
> A scattered failure pattern like this — one small, one large, the rest centred — points to panel-level variation (cutting or sewing inconsistency) rather than a shifted process: if all ten had read high, the pattern or marker would be suspect. Measurement data is diagnostic, not just judgmental: the auditor reports the pattern of failures, not only the count.

4. Final audit, needle control and carton audit

The **final audit** applies the ISO 2859-1 chain of AM-3207 Topic 9 to packed lots (course standard: lot 2,400 → code K → n = 125 → majors Ac 7/Re 8 at AQL 2.5, minors Ac 10/Re 11 at AQL 4.0, criticals zero). **Needle control** is a documented system — broken-needle procedure, needle inventory, and a conveyor **metal detector** through which every garment passes after finishing; typical conveyor sensitivity is quoted against a standard ferrous test sphere (about 1.2 mm) and the machine is challenge-tested with test cards at set intervals, with garments segregated until a passed re-run. The **carton audit** finally opens a sample of sealed cartons to verify count, size/colour assortment, labelling and packing standard against the packing list — the last chance to catch a logistics error before it becomes a claim.

Glossary

TERM	MEANING
In-line inspection	Checking at the operation during production, where correction is cheapest.
End-of-line inspection	100% check of finished garments before finishing and packing.
Points of measurement	The defined locations and methods by which garment dimensions are audited against specification and tolerance.

Needle control	The documented system for needle issue, broken-needle recovery and metal detection of finished garments.
Carton audit	Opening sampled sealed cartons to verify count, assortment, labelling and packing.

Key points and summary

- Quality is protected by layers — in-line, end-of-line, measurement audit, final AQL audit, metal detection, carton audit — each catching what the previous cannot.
- In-line DHU converts inspection into a floor signal (worked: 6.0 at the roving check).
- Measurement audit judges values against tolerance and reads their pattern: 2 of 10 chests out of tolerance, scattered — a panel-variation signal.
- The final audit reuses the established AQL chain; metal detection and carton audit close the safety and logistics gates.

Practice

Answer in your own words; brief model answers follow.

1. A specification gives sleeve length 62 cm ± 1.0 cm. Measured: 61.4 · 62.2 · 60.8 · 61.9 · 62.6. Which garments fail?
2. An in-line check of 400 garments finds 22 defects. Calculate the DHU.
3. Why is metal detection placed after finishing rather than at end-of-line inspection?
4. Ten chest measurements all read between 113.6 and 114.2 cm against the 112 ± 1.5 cm specification. What does this pattern indicate, differently from the worked example?

Answer key

1. Acceptable range 61.0–63.0 cm. Only 60.8 fails (below 61.0). The other four pass.
2. DHU = 22 ÷ 400 × 100 = 5.5.
3. Metallic contamination can be introduced at any stage up to the last handling — pressing, trimming, packing tools. Detection must gate the final state of the garment, so it stands after every operation that could add metal.
4. All values are high and tightly grouped — a systematic shift, not scatter: suspect the pattern, the marker, or shrinkage compensation (AM-3207 Topic 2), and correct the source rather than sorting garments.